

Syllabus - General Astronomy

Spring Semester, 2026

Course Title	General Astronomy	Course Code		PE3201	Credit	3	Classroom	Tamgu 2403
		Type		Elective	Hours	3		
Teacher	Name	Class no.	Office	Office hours	Contact		Google Classroom	
	PARK Sun Mie	#1, #2	Changjo #5504	Mon. 13:40-16:30 Wed. 14:40-15:30	smpark88@gmail.com 051-606-2241		pgrfsl72	

1. Course Descriptions

Course Objectives	
This course provides a comprehensive introduction to the universe as the totality of space, time, matter, and energy. Topics include the solar system, stars, galaxies, and the foundations of modern cosmology. In addition to examining the structure and origins of astronomical objects, the course introduces how astronomers collect and analyze observational data.	
Contents	
This course offers a comprehensive overview of modern astronomy. It covers the origins and development of modern astronomy, the solar system, stars, galaxies, and the origin and evolution of the universe.	

2. Core Target Competencies

Subject Core Competencies					
Ability to understand and apply scientific knowledge	Scientific thinking	Problem-solving skills	Ability to explore and experiment	Creativity	Ability to collaborate and communicate

3. Textbooks & References

- A. Text : Horizons: Exploring the universe, 15E, M. Seeds and D. Backman (Brooks/Cole, 2025)
- B. References: Foundations of Astronomy, 14E, Michael A. Seeds, Dana Backman (Cengage Learning, 2018)
- Astronomy: A Beginner's Guide to the Universe, 8E, Eric Chaisson, Steve McMillan (Pearson, 2016)
- The Essential Cosmic Perspective, 7E, J. Bennett, M. Donahue, N. Schneider, M. Voit (Pearson, 2015)

4. Class Method

- A. Presentation and discussion
- B. Term Project

5. Evaluation : The grade will be given according to total points.

A. Evaluation Criteria

Presentation & Homework	Term Project	Midterm Exam	Final Exam	Attitude and Attendance	Total (%)
20	10	30	30	10	100

B. Evaluation Method

(1) Presentation (Group/Individual evaluation)

- You should present one among presentation topics in each chapter.
- Evaluation standards : ① They present logically, interestingly and confidently?
② During the task, their roles are well organized? ③ They try to study creatively?
④ There is any further investigation? ⑤ They make good use of figures and tables?
- 10-min presentation plus 5-min questions and discussion.
- Textbook and other materials can be used.

(2) Homework (Individual evaluation) : See the lecture schedules.

- In time you will get the score range of A to D, and it will be normally 1 week and the exact deadline will be announced in the classroom. If you submit your report after that but within 1 week, then you will get a score degraded by 10% each day. After 1 week, any report cannot be accepted and the score is automatically F. If you have any problem, please consult with the instructor in advance.

* **Plagiarism** : A meeting of faculty committee will be open for FAIL of the course.

(3) Term Project (Group/Individual evaluation)

- You should study a topic related to this class for a semester and present the results.
- It is necessary to select a **scientifically meaningful topic** beyond simply having fun.
- Evaluation standards : ① research content, ② presentation, and ③ report
- The research content is assessed through the ① plan, ② process, and ③ result.
- The creativity, scientific logic, and cooperation will be checked.
- The presentation follows the above evaluation criteria.
- The report is assessed based on the completeness (introduction, method, result, and discussion).

(4) Attitude (Individual evaluation)

- They participate actively and interestingly in class?
- They think and ask a lot of questions?

(5) Attendance

- Unexcused absence makes the score degraded by 10% each time, except illness (with confirmation from school infirmary and contact to instructor in advance) or official cases.

6. Remarks

A. **14 kinds of topics** about astronomy will be conducted in this semester.

I will upload all lecture files on Google classroom.

B. Presentation

- I will share some of the topic per chapter in advance.
- Only 10 minutes amount for slide presentation is allowed per each group.
Do not spend much time just for decorating.
- Textbook and other materials can be used freely.
- After the presentation, we will have discussion time.

You need to

- Actively, interestingly participate in class.
- Think and ask a lot of questions to me and yourself.

7. Criteria for Awarding Credits

 Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees : The standard score (Z) from the taken examination will be converted into the corresponding standard score of the untaken examination.
- If a basis score cannot be calculated, the grading method : Article 10, Paragraphs 2 and 3 of the "Academic Performance Management Regulations" will apply accordingly.

※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

8. Weekly Plans

Week	Topic and Contents	(Notes)
Week 01 2.23.~3.1.	Introduction to the syllabus Ch01. Here and Now	R. 4, 5, 7
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills	
Week 02 3.2.~3.8.	Ch02. A Guide to the Sky	R. 6, 11, 16 3.2. Substitute Holiday
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills	
Week 03 3.9.~3.15.	Ch03. Cycles of the Sun and Moon	R. 3, 8, 9
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 04 3.16.~3.22.	Ch04. Origin of Modern Astronomy	R. 3, 9, 12
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 05 3.23.~3.29.	Ch06. Atoms and Spectra	R. 6, 7 / P. 1, 7
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 06 3.30.~4.5.	Ch07. The Sun	Determination of topic for term project(TP) R. 8, 10, 14
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Ability to explore and experiment - Creativity - Ability to collaborate and communicate	
Week 07 4.6.~4.12.	Ch08. The Family of Stars	R. 5, 9 / P. 8, 12
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	

Week 08 4.13.~4.19.	Mid-term Examination (4.14.~4.17.)	4.13. Regular classes
Week 09 4.20.~4.26.	Ch09. The Formation of Stars and Planets	R. 2, 8, 9
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 10 4.27.~5.3.	Ch10. Stellar Structure and Evolution	R. 3, 9, 18
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 11 5.4.~5.10.	Ch11. Supernova, Neutron Stars, and Black Holes	R. 1, 10, 16 / P. 8 5.5. Children's Day
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 12 5.11.~5.17.	Ch12. The Milky Way Galaxy	Interim presentation of TP R. 3, 7, 14
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Ability to explore and experiment - Creativity - Ability to collaborate and communicate	
Week 13 5.18.~5.24.	Ch13. Galaxies: Normal and Active	R. 1, 5, 10 / P. 4
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 14 5.25.~5.31.	Presentation of term project result	5.25. Substitute Holiday
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Ability to explore and experiment - Creativity - Ability to collaborate and communicate	
Week 15 6.1.~6.7.	Ch14. Modern Cosmology	R. 4, 7, 11 / P. 7 6.3. Local Election Day
Core Competencies	- Ability to understand and apply scientific knowledge - Scientific thinking - Problem-solving skills - Creativity - Ability to collaborate and communicate	
Week 16 6.8.~6.14.		Make-up class week
Week 17 6.15.~6.19.	Final Examination (6.15.~6.19.)	

※ The above plans may change according to the academic schedule.

Presentation topics for Week 03

- Orbital motion of the moon and cycle of phases
- Lunar and solar eclipses
- Why do tides happen?
- Origin of the Moon and interior of the Moon

For presentation, each group choose one thing among above topics.

Of course, this topic is selected in each chapter.

I will upload the topics for presentation on Mondays through the Google classroom "General Astronomy".

I suggest that you check the topics and select one after discussion time in your group.

The presentation will be a 10 min talk for each group.

After the presentation, we will have discussion time.